

Project Design Phase

Solution Architecture

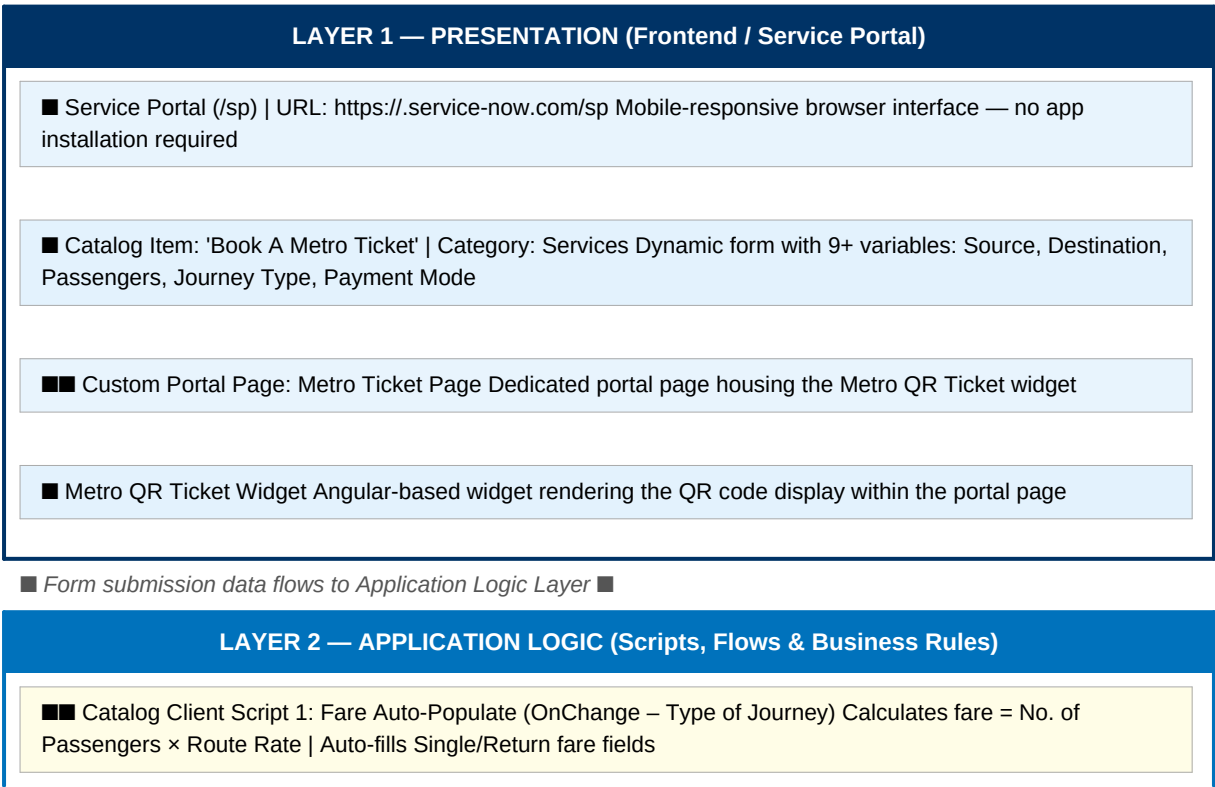
Date	15 February 2025
Team ID	HYD-METRO-2025
Project Name	Hyderabad Metro Rail – Digital Ticketing System (ServiceNow)
Maximum Marks	4 Marks

Solution Architecture Overview

Solution architecture bridges the gap between the business problem (manual metro ticketing) and the technology solution (ServiceNow Digital Ticketing System). The architecture below describes the structure, components, behavior, and data flows of the complete system.

Architecture Goals
- Find the best tech solution: ServiceNow SaaS platform — zero infrastructure, rapid deployment - Describe structure & behavior: 3-tier architecture (Portal → Logic → Data) detailed below - Define features & phases: 7 sprints from ideation to UAT, all components configured - Provide specifications: Component table, data flow, ACL matrix, and script inventory included

Solution Architecture Diagram



■ Catalog Client Script 2: QR Code Generator (OnSubmit) Triggers GlideModal with QR URL:
<https://example.com/ticket?id=>

■ Catalog Client Script 3: Field Validator (OnChange – No. of Passengers) Validates numeric input | Shows alert for non-numeric values

■ Catalog UI Policy: Conditional Field Display Condition: Mode of Payment = 'Others' → Shows 'Enter Payment Mode' text field

■ Flow Designer: Metro Project Flow Trigger: Service Catalog (RITM) | Action 1: Get Catalog Variables | Action 2: Create Record in Metro Database

■ Business Rule: Generate Metro Ticket Fires on Metro Database insert | Auto-generates ticket reference | Triggers email notification

■ ACL Rules (Role-Based Access Control) Commuters: submit only via catalog | station_manager: read Metro DB | admin: full CRUD

■ Processed data flows to Data Layer ■

LAYER 3 — DATA (Custom Tables & ServiceNow System Tables)

■ Custom Table: Metro Station's Details Fields: Station Name | Station Code | Line | Zone | Address | Active Purpose: Referenced by catalog Source/Destination dropdowns (Reference field type) Records: 10+ Hyderabad Metro stations pre-populated

■ Custom Table: Metro Database Fields: Passenger Name | Starting From (ref) | Going To (ref) | Journey Type | No. of Passengers Fare Amount | Mode of Payment | Booking Date | Ticket Reference Purpose: Stores every booking transaction | Protected by ACLs

■ ServiceNow System Tables (Used by solution) sc_cat_item (Catalog Items) | sc_item_option (Variables) | sc_req_item (RITM) sys_user (Commuter records) | sys_security_acl (ACL rules) | sysevent_email_action (Notifications)

EXTERNAL SYSTEMS & INFRASTRUCTURE

■ Email System (ServiceNow Notification Engine) Sends HTML booking confirmation email to commuter's email address on RITM submission

■ External QR API (external_qr_api_page) Called by GlideModal to render QR code from booking URL — currently using dummy endpoint Production: integrate with HMRL gate validation API

■■ ServiceNow Cloud Infrastructure Hosted on ServiceNow multi-region SaaS cloud | 99.9%+ uptime SLA
HTTPS enforced | Session management | Multi-tenant data isolation

End-to-End Data Flow

Step	Actor	Action	System Component	Output
1	Commuter	Opens browser, navigates to /sp	ServiceNow Service Portal	Portal home page loads
2	Commuter	Searches and opens 'Book A Metro Service Catalog (sc_cat_item)	Service Catalog (sc_cat_item)	Catalog form renders with all variables
3	Commuter	Selects Source & Destination stations	Reference variable → Metro Station Details	Station names shown from master table
4	Commuter	Enters No. of Passengers	Client Script 3 (OnChange)	Validates numeric; alert if invalid
5	Commuter	Selects Journey Type (Single/Return)	Client Script 1 (OnChange)	Fare auto-populated in correct field
6	Commuter	Selects Payment Mode = 'Others'	UI Policy	Enter Payment Mode field becomes visible
7	Commuter	Clicks Submit	Client Script 2 (OnSubmit)	GlideModal opens with QR code displayed
8	ServiceNow	RITM record created	sc_req_item table	Requested Item (RITM) generated
9	ServiceNow	Flow Designer trigger fires	Flow: Metro Project	Flow execution begins
10	ServiceNow	Get Catalog Variables action	sc_item_option (Flow Action 1)	All variable values extracted
11	ServiceNow	Create Record action	Metro Database (Flow Action 2)	New booking record created in DB
12	ServiceNow	Business Rule fires on DB insert	Business Rule: Generate Metro Ticket	Ticket reference generated
13	ServiceNow	Email notification sent	Notification Engine	Confirmation email delivered to commuter
14	Station Manager	Views Metro Database table	ACL-protected custom table	Booking records visible to authorized role

Full Component Inventory

#	Component	Type	Technology	Purpose
1	Book A Metro Ticket	Service Catalog Item	ServiceNow Catalog	Primary user booking interface
2	Metro Station's Details	Custom Table	ServiceNow Custom Table	Metro station master data store
3	Metro Database	Custom Table	ServiceNow Custom Table	Booking transaction records
4	Fare Auto-Populate Script	Catalog Client Script	JavaScript / GlideForm API	Auto-calculate fare on route change
5	QR Code Generator	Catalog Client Script	JavaScript / GlideModal	Display QR on form submission
6	Field Validator	Catalog Client Script	JavaScript / GlideForm API	Validate numeric passenger count
7	Payment Mode Policy	Catalog UI Policy	ServiceNow UI Policy	Show/hide payment mode field
8	Metro Project	Flow Designer Flow	ServiceNow Flow Designer	Post-submission automation
9	Generate Metro Ticket	Business Rule	Server-side JS / GlideRecord	Ticket ID generation + notification
10	Metro Database ACLs	Access Control Rules	ServiceNow ACL Engine	Role-based data security
11	Metro Ticket Page	Service Portal Page	ServiceNow Service Portal	Custom portal page for commuters
12	Metro QR Ticket Widget	Service Portal Widget	AngularJS / HTML	QR display widget on portal page
13	Booking Confirmation Email	Notification	ServiceNow Email Engine	Confirm booking to commuter